TO:	Head of Operations, Los Angeles Region	**FROM:**	Urban Intelligence & Data Science Team
RE:	Graph ML Traffic Forecasting — Deployment Readiness	**REF:**	EM627 — METR-LA Traffic Intelligence Project

WHAT THE MODEL DOES

We have developed and validated a **Graph Neural Network (T-GCN)** traffic forecasting system trained on 4 months of real highway sensor data from 207 loop detectors across Los Angeles. The model predicts traffic speeds at all 207 sensor locations simultaneously at **5, 15, and 30 minutes ahead**, updating every 5 minutes.

0.140	0.200	5.2%	0.672
MAE at 5-min (best model)	MAE at 15-min (beats RF baseline)	Improvement over best non-graph model	Spatial correlation (justifies graph model)

WHY IT MATTERS

Current dispatch systems react to congestion *after* it forms. This model predicts congestion **15–30 minutes before it peaks**, enabling proactive repositioning of drivers before demand-supply mismatch occurs. Analysis of 34,255 traffic snapshots shows that **22.8% of highway observations involve congested conditions** — exactly when accurate ETAs and driver availability matter most to customers.

Key finding: Sensors 16 and 196 (near Downtown LA, coordinates 34.07°N 118.23°W) are congested **100% of observed time**. These two zones alone affect thousands of daily pickups and require permanent elevated driver staffing — not just reactive surge pricing.

WHERE ACTION IS NEEDED

Priority	Zone	Condition	Recommended Action
CRITICAL	Sensors 16, 196 (34.07°N 118.23°W)	Congested 100% of time	Permanent driver buffer — 5 extra drivers at all times
CRITICAL	Sensor 56 (34.09°N 118.24°W)	Congested 91% of time	Pre-position drivers by 6:45am and 4:45pm daily
HIGH	58 high-congestion sensors	Congested 25–75% of time	Surge pricing trigger when predicted speed below −0.3
HIGH	42 low-reliability sensors	MAE > 0.245 (top 20%)	Display ETA uncertainty flag in customer app
STANDARD	All 207 sensors	Evening rush (5–7pm) more severe than morning	Increase driver incentives from 4:30pm Mon–Fri

DECISION RECOMMENDATION

The T-GCN model is ready for **pilot deployment** as a decision-support layer for the LA operations team. It should be integrated into the dispatch dashboard to flag predicted congestion zones 15 minutes in advance. For 30-minute predictions, add a 15% ETA buffer to customer-facing estimates to account for reduced model accuracy at longer horizons.

Immediate next step: Retrain the model on 2024–2025 data to capture post-pandemic commuting pattern shifts. Current model was validated on 2012 sensor data; performance on modern traffic patterns requires validation before full production deployment.